

26.4.a The the numerator and denominator share no common factor and the denominator is 0 at $0, i, -i$, so f is not defined at these points. By the quotient rule, the derivative exists at all other points. So $0, i, -i$ are singular points and the function is analytic everywhere else.

26.6 It's simplest to use the polar form of the Cauchy-Riemann equations. We see that $u_r = 1/r$, $u_\theta = 0$, $v_r = 0$ and $v_\theta = 1$. So the polar Cauchy-Riemann equations are satisfied and the partials are continuous. Thus the function is defined on the domain (notice that the domain is carefully chosen so that θ is well defined without ambiguity).

Now, from our geometrical understanding of functions, z^2 takes the open first quadrant to the upper half plane and adding one then shifts everything one unit to the right. So, as the suggestion notes, $\text{Im}(z^2 + 1) > 0$ for z in the open first quadrant, i.e. $0 < \text{Arg}(z^2 + 1) < \pi$. Since this is in the domain of g , the composition rule tells us that $g(z^2 + 1)$ is analytic.

The rest follows from the CR formula for the derivative of g and from the chain rule.

26.7 If f is real valued and analytic on D , then $f = \bar{f}$, so $\bar{f}$ is also analytic on D . The result now follows from Example 3 in Section 26.

27.2 The formulas $u(x, y) = c_1$ and $v(x, y) = c_2$ determine curves that locally take the for $y = y(x)$. As noted in the suggestion, we have from the multivariable chain rule (differentiating $u(x, y)$ with respect to x) that $u_x + u_y \frac{dy}{dx} = 0$ and $v_x + v_y \frac{dy}{dx} = 0$.

Find all real values of a, b, c, d so that $ax^3 + bx^2y + cxy^2 + dy^3$ is harmonic. Let $u = ax^3 + bx^2y + cxy^2 + dy^3$. Then $u_x = 3ax^2 + 2bxy + cy^2$ and $u_{xx} = 6ax + 2by$. Similarly, $u_y = bx^2 + 2cxy + 3dy^2$ and $u_{yy} = 2cx + 6dy$. So u is harmonic if and only if $6a = -2c$ and $2b = -6d$, i.e. $c = -3a$ and $b = -3d$. So any function of the form $u = ax^3 - 3dx^2y - 3axy^2 + dy^3$ is harmonic.

Show by hand that $u = x^3 - 3x^2y - 3xy^2 + y^3$ is harmonic. Find a v so that $f = u + iv$ is entire (hint: use the Cauchy-Riemann equations). $u_x = 3x^2 - 6xy - 3y^2$ and $u_{xx} = 6x - 6y$. $u_y = -3x^2 - 6xy + 3y^2$ and $u_{yy} = -6x + 6y$. So u is harmonic.

By the Cauchy-Riemann equations, to find our desired v we need $v_x = -u_y = 3x^2 + 6xy - 3y^2$ and $v_y = u_x = 3x^2 - 6xy - 3y^2$. Integrating v_x with respect to x , we see that we must have $v = x^3 + 3x^2y - 3xy^2 + g(y)$ for some function g depending only on y . Taking the y derivative of this, we must have $v_y = 3x^2 - 6xy + \frac{dg}{dy}(y)$. Comparing with the v_y we must have from the Cauchy-Riemann equations, we see that $\frac{dg}{dy}(y) = -3y^2$. So $g(y) = -y^3 + C$. So we see that $v = x^3 + 3x^2y - 3xy^2 - y^3 + C$ satisfies the Cauchy-Riemann equations with u , and everything in sight is continuous. So any such v works.

Homework 6

30.1.b $e^{\frac{2+\pi i}{4}} = e^{1/2}e^{\pi i/4} = \sqrt{e}(\cos \pi/4 + i \sin \pi/4) = \sqrt{e}(1/\sqrt{2} + i/\sqrt{2}) = \sqrt{e/2}(1 + i)$

30.3 $e^{\bar{z}} = e^x(\cos y - i \sin y)$. So $u_x = e^x \cos y$, $u_y = -e^x \sin y$, $v_x = -e^x \sin y$, $v_y = -e^x \cos y$. If CR holds, then $\cos y = -\cos y$, so $\cos y = 1/2$ and $\sin y = -\sin y$, so $\sin y = 1/2$. Since $\sin y$ and $\cos y$ are never both $1/2$ for the same y , $e^{\bar{z}}$ cannot be differentiable anywhere.

30.6 $|e^{z^2}| = |e^{x^2-y^2+i2xy}| = |e^{x^2-y^2}| |e^{i2xy}| = e^{x^2-y^2} = e^{x^2+y^2} e^{-y^2} = e^{|z|^2} e^{-y^2}$. Since $-y^2 \leq 0$ we have $e^{-y^2} \leq 1$. So $|e^{z^2}| \leq e^{|z|^2} e^{-y^2}$.

30.8.a $e^z = -2$ implies that $e^x(\cos y + i \sin y) = -2$. Since -2 is real, $\sin y = 0$, so y is a multiple of π . Since $-2 < 0$, we must have $\cos y < 0$, which means that is an odd multiple of π , i.e. $y = (2n+1)\pi$. Then $e^z = -e^x = -2$, so $e^x = 2$, which implies that $x = \ln 2$.

30.11 As $x \rightarrow -\infty$, e^z moves toward 0 along a ray from the origin. As $y \rightarrow \infty$, e^z moves clockwise around a circle of radius e^x centered at the origin an infinite number of times.

38.7 for $|\sin z|^2$ Using (13), $|\sin z|^2 = \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y = \sin^2 x \cosh^2 y - \sin^2 x \sinh^2 y + \sin^2 x \sinh^2 y + \cos^2 x \sinh^2 y = \sin^2 x (\cosh^2 y - \sinh^2 y) + \sinh^2 y (\sin^2 x + \cos^2 x) = \sin^2 x + \sinh^2 y$.

Show that $\sin^2 z + \cos^2 z = 1$ $\sin^2 z + \cos^2 z = \left(\frac{e^{iz} - e^{-iz}}{2i} \right)^2 + \left(\frac{e^{iz} + e^{-iz}}{2} \right)^2 = \frac{e^{2iz} + e^{-2iz} - 2}{-4} + \frac{e^{2iz} + e^{-2iz} + 2}{4} = 4/4 = 1$

Show that $\overline{e^z} = e^{\bar{z}}$: $\overline{e^z} = \overline{e^x \cos y + i e^x \sin y} = e^x \cos y - i e^x \sin y = e^x \cos(-y) + i e^x \sin(-y) = e^x e^{-iy} = e^{x-iy} = e^{\bar{z}}$.

Show that $\overline{\sin z} = \sin \bar{z}$ and $\overline{\cos z} = \cos \bar{z}$. Using the last problem and basic properties of conjugates, $\overline{\sin z} = \frac{e^{-i\bar{z}} - e^{i\bar{z}}}{-2i} = \frac{e^{i\bar{z}} - e^{-i\bar{z}}}{2i} = \sin \bar{z}$. $\cos$ is similar.

33.1.a $|-ei| = e$ and $\arg(-ie) = -\pi/2 + 2\pi n$, so $\text{Log}(-ei) = \ln e + i\theta = 1 + i(-\pi/2)$

33.2.c $|-1 + \sqrt{3}i| = 2$ and $\arg(-1 + \sqrt{3}i) = 2\pi/3 + 2\pi n$, so $\log(-1 + \sqrt{3}i) = \ln 2 + i(2\pi/3 + 2\pi n)$

33.4 With the chosen branch, $\log(i^2) = \log(-1) = \pi i$, while $2 \log i = 2(i5\pi/2) = i5\pi$

33.9 In general, we have $e^z = e^{x+iy} = e^x e^{iy}$, so $|e^z| = e^x$ and $\arg(e^z) = y + 2\pi n$. If we use the branch $\alpha < \theta < \alpha + 2\pi$ for $\log$, then we have that $\log(e^z) = \ln e^x + i\Theta$, where Θ is the value of $\arg(e^z) = y + 2\pi n$ with $\alpha < \Theta < \alpha + 2\pi$. But with the assumption, this is precisely y . So $\log e^z = x + iy = z$.

33.10.a By definition/branch cuts, $\text{Log}(z)$ is analytic so long as z does not lie on the part of the real axis with $x \leq 0$. Thus, since the composition of analytic functions is analytic where it is defined (and $\text{Log } z$ and $z - i$ are analytic) $\text{Log}(z - i)$ is thus analytic so long as $z - i$ is not on the non-positive x -axis, which is equivalent to z not being on the line $y = 1$ with $x \leq 0$.

33.11 One way to do this is by direct computation of derivatives. Alternatively, $\ln(x^2 + y^2)$ is the real part of $2\log z$ for any branch cut. For any branch cut, $2\log z$ is analytic in its domain, and so its real part is harmonic in that domain, which covers all of the plane except one ray. If we choose a different branch cut, then we see that $\ln(x^2 + y^2)$ is also harmonic on the ray, except for at 0. Since being harmonic is a local property and we have verified it at all non-zero points of the plane, the function is harmonic on $\mathbb{C} - \{0\}$.

Homework 7

34.3 Let $z_1 = -\sqrt{2}/2 + i\sqrt{2}/2$, and let $z_2 = -i$. Then $z_1/z_2 = -\sqrt{2}/2 - i\sqrt{2}/2$, so $\text{Log}(z_1/z_2) = -i3\pi/4$. But $\text{Log}(z_1) - \text{Log}(z_2) = i3\pi/4 - (-i\pi/2) = i5\pi/4$.

36.1.a $(1+i)^i = e^{i\log(1+i)} = e^{i(\ln\sqrt{2} + i(\pi/4 + 2\pi n))} = e^{-\pi/4 + 2\pi n} e^{i\ln\sqrt{2}} = e^{-\pi/4 + 2\pi n} e^{i(\ln 2)/2}$

36.2.a Principal value of $(-i)^i = e^{i\text{Log}(-i)} = e^{i(-\pi i/2)} = e^{\pi/2}$

36.2.c Principal value of $(1-i)^{4i} = e^{4i\text{Log}(1-i)} = e^{4i(\ln\sqrt{2} - \pi i/4)} = e^{\pi + i2\ln(2)} = e^{\pi}(\cos(2\ln 2) + i\sin(2\ln 2))$

36.6 If a is real then $|z^a| = |e^{a\log z}| = |e^{a(\ln|z| + ai\arg(z))}| = |e^{a\ln|z|} e^{ai\arg z}| = |e^{a\ln|z|}| = |e^{\ln|z|^a}| = ||z|^a|$. But if we take the principal value of $|z|^a$ then since $|z|$ is a positive real number we have $|z|^a = e^{a\text{Log}|z|} = e^{a\ln|z|}$ as *real* numbers, and this is the usual real number $|z|^a$, which is also positive. So $||z|^a| = |z|^a$.

36.8.a Using principal values, we have $z^{c_1} z^{c_2} = e^{c_1 \text{Log } z} e^{c_2 \text{Log } z}$. As noted in Section 35, when we use principal values e^z is exactly the function we're used to working with, so, in particular, $e^{c_1 \text{Log } z} e^{c_2 \text{Log } z} = e^{c_1 \text{Log } z + c_2 \text{Log } z} = e^{(c_1 + c_2) \text{Log } z}$, which is the principal branch of $z^{c_1 + c_2}$.

42.2.a $\int_0^1 (1+it)^2 dt = \int_0^1 1 + 2it - t^2 dt = t + it^2 - t^3/3|_0^1 = 1 + i - 1/3 - (0) = \frac{2}{3} + i$

42.2.c $\int_0^{\pi/6} e^{i2t} dt = \frac{e^{i2t}}{2i} \Big|_0^{\pi/6} = \frac{e^{i\pi/3} - 1}{2i} = \frac{1}{2i} \left(\frac{-1}{2} + i\frac{\sqrt{3}}{2} \right) = \frac{\sqrt{3}+i}{4}$

42.4 $\int_0^\pi e^{(1+i)x} dx = \frac{e^{(1+i)x}}{1+i} \Big|_0^\pi = \frac{e^{(1+i)\pi} - 1}{1+i} = \frac{-e^\pi - 1}{1+i} = \frac{1}{2}(-e^\pi - 1)(1-i)$. So $\int_0^\pi e^x \cos x dx = \frac{1}{2}(-e^\pi - 1)$ and $\int_0^\pi e^x \sin x dx = \frac{1}{2}(e^\pi + 1)$.

43.1.a Let $\tau = -t$. Then $d\tau = -dt$ and $\int_{-b}^{-a} w(-t)dt = -\int_{-b}^{-a} w(\tau)d\tau = \int_a^b w(\tau)d\tau$

46.1.a $z' = 2ie^{i\theta}$ so $\int_C \frac{z+2}{z} dz = \int_0^\pi \frac{2e^{i\theta}+2}{2e^{i\theta}} 2ie^{i\theta} d\theta = \int_0^\pi 2ie^{i\theta} + 2i d\theta = 2e^{i\theta} + 2i\theta \Big|_0^\pi = -2 + 2\pi i - 2 = -4 + 2\pi i$

46.2.b Take $z(t) = t$, $0 \leq t \leq 2$. Then $\int_C z - 1 dz = \int_0^2 t - 1 dt = t^2/2 - t \Big|_0^2 = 2 - 2 - 0 = 0$

46.4 Our parametrization is $z = t + it^3$ but we need to consider two pieces, $-1 \leq t \leq 0$ and $0 \leq t \leq 1$. We also have $z' = 1 + 3t^2i$. The first part gives us $\int_{-1}^0 1(1 + 3t^2i) dt = t + t^3i \Big|_{-1}^0 = -(-1-i) = 1+i$. The second part is $\int_0^1 4t^3(1 + 3t^2i) dt = \int_0^1 4t^3 + 12t^5i dt = t^4 + 2t^6i \Big|_0^1 = 1 + 2i$. So altogether we have $1 + i + 1 + 2i = 2 + 3i$

46.5 Given any $z(t)$, $a \leq t \leq b$, we have $\int_C 1 dz = \int_a^b \frac{dz}{dt} dt = z|_a^b = z(b) - z(a) = z_2 - z_1$

46.6 Note that the contour only has an endpoint on the branch cut. So we use $z' = ie^{i\theta}$ and the integral is $\int_0^\pi e^{i \operatorname{Log} z} ie^{i\theta} d\theta = \int_0^\pi e^{i(i\theta)} ie^{i\theta} d\theta = \int_0^\pi e^{-\theta} ie^{i\theta} d\theta = \int_0^\pi e^{i\theta - \theta} i d\theta = \int_0^\pi e^{(i-1)\theta} i d\theta = \frac{ie^{(i-1)\theta}}{i-1} \Big|_0^\pi = \frac{ie^{\pi i - \pi} - i}{i-1} = \frac{-ie^{-\pi} - i}{i-1} = \frac{(-1-i)(-ie^{-\pi} - i)}{2} = \frac{(-1+i)(e^{-\pi} + 1)}{2}$

Homework 8

43.5 Let $w(t) = f(z(t)) = u(x(t), y(t)) + iv(x(t), y(t))$. Then by the chain rule, $\frac{dw}{dt} = \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} + i \frac{\partial v}{\partial x} \frac{dx}{dt} + i \frac{\partial v}{\partial y} \frac{dy}{dt}$. Writing $u_x, u_y, v_x, v_y, x', y'$ for the appropriate derivatives and using Cauchy-Riemann, this is $u_x x' + u_y y' + i(v_x x' + v_y y') = u_x x' - v_x y' + i(v_x x' + u_x y') = (u_x + iv_x)(x' + iy') = \frac{df}{dz} \frac{dz}{dt}$ as desired.

47.1.a On part of the circle of radius 2, $|z+4| \leq |z|+4 = 2+4 = 6$ and $|z^3-1| \geq ||z|^3-1| = 7$. The length of the quarter circle is $2\pi \cdot 2/4 = \pi$. So by the Theorem in this section, the integral is $\leq 6\pi/7$.

47.2 As noted in the suggestion, the minimum $|z|$ on the curve occurs at the midpoint $1/2 + i/2$ and so is $\sqrt{1/2}$. Thus $|1/z^4| \geq 1/(\sqrt{1/2})^4 = 4$. The length of the line segment is $\sqrt{2}$ by the Pythagorean theorem. So the integral is $\leq 4\sqrt{2}$.

47.4 On the circle of radius R we have that $|2z^2-1| \leq 2|z|^2+1 = 2R^2+1$ while $|z^4+5z^2+4| = |z^2+4||z^2+1| \geq ||z|^2-4||z|^2+1| = (R^2-4)(R^2+1)$. So $|\frac{2z^2-1}{z^4+5z^2+4}| \leq \frac{2R^2+1}{(R^2-4)(R^2+1)}$ and the integral is $\leq \frac{2R^2+1}{(R^2-4)(R^2+1)}\pi R$. As $R \rightarrow \infty$ this expression goes to 0.

47.7 $|x + i\sqrt{1-x^2} \cos \theta| = \sqrt{x^2 + (1-x^2) \cos^2 \theta}$. Since $1-x^2 \geq 0$ by the assumption $|x| \leq 1$, we have $(1-x^2) \cos^2 \theta \leq 1-x^2$. So $|x + i\sqrt{1-x^2} \cos \theta| \leq x^2 + 1 - x^2 = 1$. So $|(x + i\sqrt{1-x^2} \cos \theta)^n| = |x + i\sqrt{1-x^2} \cos \theta|^n \leq 1^n = 1$. So the modulus of the integral is $\leq \pi$ and therefore $|P_n(x)| \leq 1$.

49.1 For n a nonnegative integer, z^n is entire with antiderivative $F(z) = \frac{z^{n+1}}{n+1}$ in the whole plane. So a contour integral from z_1 to z_2 of z^n is $F(z_2) - F(z_1) = \frac{z_2^{n+1} - z_1^{n+1}}{n+1}$.

49.2.b $2 \sin(z/2)$ is an antiderivative of $\cos(z/2)$. So the integral is $2(\sin(\pi/2+i) - \sin(0)) = 2 \sin(\pi/2+i) = \frac{e^{i(\pi/2+i)} - e^{-i(\pi/2+i)}}{i} = \frac{e^{-1+i\pi/2} - e^{1-i\pi/2}}{i} = \frac{e^{-1}i - e(-i)}{i} = e^{-1} + e$

49.3 Since the case $n = 0$ is omitted, $(z - z_0)^{n-1}$ has antiderivative $\frac{(z-z_0)^n}{n}$ on the domain $\mathbb{C} - \{z_0\}$ (if $n > 0$ this is an antiderivative on all of $\mathbb{C}$). As long as a contour does not pass through z_0 , the integrand is defined and continuous on the contour. So why the Theorem in Section 48, the integral is 0 when the contour is also closed.

49.4 As observed in the text, the branch of $z^{1/2}$ using $\pi/2 < \theta < 5\pi/2$ is defined at all points of any contour that lies below the real axis except for its endpoints at -3 and 3 , and for all points on such a contour the values of this branch agree with those of the branch $0 < \theta < 2\pi$, except at the point 3 . But as one point does not affect the value

of the integral, we can use the $\pi/2 < \theta < 5\pi/2$ branch to compute. Furthermore, this branch of $z^{1/2}$ has antiderivative $z^{3/2}/(3/2)$ (with the same branch choice) on a domain containing the contour. So by the Theorem in Section 48, the contour integral is $\frac{2}{3}(3^{3/2} - (-3)^{3/2})$. For this branch, $3^{3/2} = (3e^{2\pi i})^{3/2} = 3^{3/2}e^{3\pi i} = -3^{3/2}$ while $(-3)^{3/2} = (3e^{\pi i})^{3/2} = 3^{3/2}e^{i3\pi/2} = -i3^{3/2}$. So the integral is $\frac{2/3}{1} - 3^{3/2} + i3^{3/2} = 2(-\sqrt{3} + i\sqrt{3})$.

49.5 As in the suggestion, if we consider the branch of z^i with $-\pi/2 \arg z < 3\pi/2$ to write $z^i = e^{i \log z}$ then this agrees with $e^{i \operatorname{Log} z}$ on the entirety of any contour that lies about the real axis (except for its endpoints). So the integral is the same treating z^i either way. But now using our branch, z^i has antiderivative $z^{i+1}/(i+1)$ (using the same branch to define z^{i+1}) and so by the Theorem in Section 48, the integral is $(1^{i+1} - (-1)^{i+1})/(i+1)$. Using our branch, we compute $1^{i+1} = e^{(i+1) \log 1} = e^{(i+1)(0+0i)} = e^0 = 1$, while $(-1)^{i+1} = e^{(i+1) \log(-1)} = e^{(i+1)(\ln 1 + i\pi)} = e^{-\pi + i\pi} = -e^{-\pi}$. So the integral is $\frac{1+e^{-\pi}}{i+1} = \frac{1+e^{-\pi}}{2}(1-i)$

Homework 9

53.1.a This function is analytic except at $z = -3$. So it is analytic on and inside the circle, so the integral is 0.

53.1.c This function only fails to be analytic at $\frac{-2 \pm \sqrt{4-8}}{2} = -1 \pm i$. These points are outside the disk, so by Cauchy-Goursat the integral is 0.

53.1.f f fails to be analytic at the points $\{x + iy | x \leq -2, y = 0\}$. These don't intersect the unit disk, so by Cauchy-Goursat the integral is 0.

53.2.a This function is analytic except where $z = \pm\sqrt{13}i$. So, for example, f is analytic on the domain $|z| > 0.6$. Thus the Theorem (more precisely the Corollary) from Section 53 applies.

53.3 $z_0 = 2 + i$ is in the interior of the rectangle described, and for all n the function $(z - 2 - i)^{n-1}$ is analytic on $\mathbb{C} - \{z_0\}$. So by the Corollary of Section 53, the integral on C_0 equals the integral on C . The result follows from the C_0 integral computations given.

53.6 We can't use Cauchy-Goursat here because the square root isn't well defined at 0, so the hypotheses of the theorem aren't satisfied. However, we compute the pieces. On the semicircle we use $z = e^{i\theta}$ so $z' = ie^{i\theta}$ and we have $\int_0^\pi e^{i\theta/2} ie^{i\theta} d\theta = i \int_0^\pi e^{i3\theta/2} d\theta = e^{i3\theta/2} 2/3 = (2/3)(e^{i3\pi/2} - 1) = (2/3)(-i - 1) = -2/3 - i2/3$. For the part on the positive x axis we have $\int_0^1 \sqrt{t} dt = (2/3)t^{3/2}|_0^1 = 2/3$. For the negative real part we have $\int_{-1}^0 \sqrt{-t} e^{i\pi/2} dt = i2/3$. Adding these three pieces, we get 0.

53.7 This comes from Green's Theorem. With $f = x - iy$, Green's Theorem (see (4) in Section 50) says that $\int_C u - iv dz = \iint_R v_x - u_y + i(u_x + v_y) dA = \iint_R 0 - 0 + i(1 + 1) dA = \iint R 2i dA = 2i \cdot \text{Area}(R)$.

Homework 10

57.1.a $2\pi i e^{-\pi i/2} = 2\pi i(-i) = 2\pi$

57.1.b $2\pi i \frac{\cos(0)}{0^2+8} = \pi i/4$

57.1.c $\frac{z}{2z+1} = \frac{z/2}{z+1/2}$ so the answer is $2\pi i(-1/4) = -\pi i/2$

57.1.e $2\pi i \frac{d}{dz} \tan(z/2) = 2\pi i \sec^2(z/2)/2$. So the answer is $\pi i \sec^2(x_0/2)$

57.2.a $z^2 + 4 = (z + 2i)(z - 2i)$. $2i$ is in the circle but $-2i$ isn't. So the integral is $2\pi i$ times $\frac{1}{z+2i}$ evaluated at $2i$. So the answer is $2\pi i/4i = \pi/2$

57.3 $g(2) = \int_C \frac{2s^2-s-2}{s-2} ds$, so $g(2) = 2\pi i(2(2)^2 - 2 - 2) = 2\pi i 4 = 8\pi i$. If $|z| > 3$ then the integrand is analytic on and inside the contour so $g(z) = 0$.

57.5 If z_0 is outside the contour, both sides are 0. If z_0 is inside the contour and C is positively oriented then from the Cauchy formulas both sides are $f'(z_0)$, using that f analytic implies that f' is analytic. If C is negatively oriented, then the integrals reverse signs but are still equal.

57.7 From the integral formula, the integral is $2\pi i e^{a0} = 2\pi i$. In terms of θ , we let $z = e^{i\theta}$ so that $z' = ie^{i\theta}$. Then the integral is $\int_{-\pi}^{\pi} e^{a(\cos\theta+i\sin\theta)} ie^{i\theta}/e^{i\theta} d\theta = i \int_{-\pi}^{\pi} e^{a\cos\theta} e^{ia\sin\theta} d\theta = i \int_{-\pi}^{\pi} e^{a\cos\theta} (\cos(a\sin\theta) + i\sin(a\sin\theta)) d\theta$. Since $e^{a\cos\theta} \sin(a\sin\theta)$ is an odd function, its integral from $-\pi$ to π is 0. So we get $i \int_{-\pi}^{\pi} e^{a\cos\theta} \cos(a\sin\theta) d\theta = 2\pi i$. Also $e^{a\cos\theta} \cos(a\sin\theta)$ is even so its integral from $-\pi$ to 0 is equal to the integral from 0 to π . So $\int_0^{\pi} e^{a\cos\theta} \cos(a\sin\theta) d\theta = \pi$

57.10 Let z_0 be a point in the plane, and let C_R be the circle of radius R around z_0 . The points on C_R have the form $z_0 + Re^{i\theta}$. Then on C_R we have $|f(z)| \leq A|z_0 + Re^{i\theta}| \leq A(|z_0| + R)$. So $f''(z_0) \leq 2A(|z_0| + R)/R^2$. As R goes to ∞ , we see that $f''(z_0) = 0$. So $f'(z)$ is constant, i.e. $f'(z) = a_1$. Since f is an antiderivative of f' , this implies $f(z) = a_1 z + C$. But $f(0)$ must be 0 for $|f(z)| \leq A|z|$ to hold, $f(z) = a_1 z$.

59.1 If f is entire, so is $e^f = e^{u+iv} = e^u e^{iv}$. So if u is bounded, then so is $|e^f| = e^u$. So then e^f is constant. So $|e^f| = e^u$ is constant. So $u = \ln e^u$ is constant.

59.3 Let R be the region $|z| \leq 1$. Then if $f(z) = z$, $|f(z)| = |z| = 0$ at $z = 0$, but $|f(z)| = |z| > 0$ for $z \neq 0$. So $|f|$ has a minimum in the interior of R .

59.8.a Just multiply and cancel

59.8.b By part a, we can write $z^k - z_0^k = (z - z_0)P_{k-1}(z)$, where P_{k-1} is a polynomial of degree $k - 1$.

$$\begin{aligned} P(z) - P(z_0) &= a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n - a_0 - a_1 z_0 - a_2 z_0^2 - \cdots - a_n z_0^n \\ &= a_1(z - z_0) + a_2(z - z_0)^2 + \cdots + a_n(z^n - z_0^n) \\ &= a_1(z - z_0) + a_2(z - z_0)P_1(z) + \cdots + a_n(z - z_0)P_{n-1}(z) \\ &= (z - z_0)(a_1 + a_2 P_1(z) + \cdots + a_n P_{n-1}(z)) \\ &= (z - z_0)Q(z) \end{aligned}$$

When $P(z_0) = 0$ we see that $P(z) = (z - z_0)Q(z)$, as required.

Suppose $f(z)$ is entire and $|f(z)| \geq 1$ for all z . Show that f is constant. Since $|f(z)| \geq 1$, we have $|\frac{1}{f(z)}| \leq 1$, so $|\frac{1}{f(z)}|$ is bounded. It is entire since $f(z)$ is never 0. So $1/f(z)$ is constant by Liouville's Theorem. So $f(z)$ is also constant.

Let R be a closed bounded region of the plane. Suppose f and g are continuous on R and analytic in the interior of R . Show that if $f = g$ on the boundary of R then $f = g$ on all of R . Consider $f - g$, which is 0 on the boundary. By the corollary to the maximum modulus principle, the maximum of $|f - g|$ is on the boundary of R , so $|f - g|$ must be 0 on all of R . So $f = g$. Technically that corollary requires $f - g$ not be constant, but if $f - g$ is constant, then since it's 0 on the boundary it's 0 everywhere in R so again $f = g$.

What is the maximum of $|e^{iz^2}|$ on the disk $|z| \leq 1$. From the Maximum Modulus Principle, we know that the maximum must be on the boundary. So we consider $|e^{iz^2}| = |e^{i(x^2 - y^2 + 2ixy)}| = e^{-2xy}$ on the circle. This will have its maximum where $-2xy$ has its maximum on the circle. Using Calc III methods (either parametrize the curve or use Lagrange multipliers), the maximum will be where $y = -x$ on the circle, i.e. $(1/\sqrt{2})(1 - i)$ and $(1/\sqrt{2})(-1 + i)$. Then $e^{-2xy} = e$.

61.1 We want $|(\frac{1}{n^2} + i) - i| = |\frac{1}{n^2}| < \epsilon$. This will be true so long as $n > 1/\sqrt{\epsilon}$

Show directly from the definitions that if $\sum_{n=1}^{\infty} a_n = A$ and $\sum_{n=1}^{\infty} b_n = B$ then $\sum_{n=1}^{\infty} (a_n + b_n) = A + B$. Let $S_m = \sum_{n=1}^m a_n$ and $T_m = \sum_{n=1}^m b_n$. Then by definition for all $\epsilon > 0$ there are M_1 and M_2 so such that $|S_m - A| < \epsilon$ for $m > M_1$ and $|T_m - B| < \epsilon$ for $m > M_2$. By taking the larger of M_1, M_2 , we see there is a single M so that $|S_m - A| < \epsilon$ and $|T_m - B| < \epsilon$ simultaneously for $m > M$. Now let $U_m = S_m + T_m = \sum_{n=1}^m (a_n + b_n)$ (we can do this because these sums are finite). Then for $M > m$, $|U_m - (A + B)| = |S_m + T_m - (A + B)| = |S_m - A + T_m - B| \leq |S_m - A| + |T_m - B|$. If we choose M so that $|S_m - A|, |T_m - B| < \epsilon/2$ for $m > M$, then we have $|U_m - (A + B)| < \epsilon$ for $m > M$. This shows that U_m converges to $A + B$ as desired.

Homework 11

65.2.b $e^z = ee^{z-1} = e \sum_{n=0}^{\infty} \frac{(z-1)^n}{n!}$. This applies for all $z \in \mathbb{C}$.

65.4 $\cos z = -\sin(z - \pi/2) = -\sum_{n=0}^{\infty} \frac{(z - \pi/2)^{2n+1}}{(2n+1)!}$. This holds everywhere in $\mathbb{C}$.

65.8.a $\cos z = \frac{e^{iz} + e^{-iz}}{2} = \frac{1}{2}(\sum \frac{(iz)^n}{n!} + \sum \frac{(-iz)^n}{n!}) = \frac{1}{2}(\sum \frac{(iz)^n}{n!} + \sum \frac{(-1)^n (iz)^n}{n!})$. Using that the terms are negatives of each other for n odd and equal to each other for n even this becomes $\frac{1}{2}(\sum \frac{2(iz)^{2n}}{(2n)!}) = \sum \frac{(iz)^{2n}}{(2n)!} = \sum \frac{i^{2n} z^{2n}}{(2n)!} = \sum \frac{(-1)^n z^{2n}}{(2n)!}$.

65.9 $f(z) = \sum_{n=0}^{\infty} (-1)^n \frac{(z^2)^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} (-1)^n \frac{z^{4n+2}}{(2n+1)!}$. Since this only has powers of the form z^{4n+2} , all of the terms $z^{4n}, z^{4n+1}, z^{4n+3}$ must have trivial coefficients, so $f^{(4n)}(0) = 0$ and similarly for the others. Notice that $4n+1$ and $4n+3$ together give all the positive odd integers, so we can restate that condition as vanishing for all $2n+1$.

Find a Maclaurin series for $\frac{z^3}{z^2+16}$. On what set does this converge? $\frac{z^3}{z^2+16} = \frac{z^3}{16} \frac{1}{1-\frac{z^2}{-16}} = \frac{z^3}{16} \sum_{n=0}^{\infty} \left(\frac{z^2}{-16}\right)^n = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+3}}{(16)^{n+1}}$. This converges when $|z^2/16| < 1$, i.e. when $|z| < 4$

Find a Taylor series for $\frac{z}{1-z}$ centered at $z = 3$. What is the region of convergence?

$$\begin{aligned}
 \frac{z}{1-z} &= \frac{z-3+3}{-2-(z-3)} \\
 &= \frac{z-3}{-2-(z-3)} + \frac{3}{-2-(z-3)} \\
 &= \frac{z-3}{-2} \frac{1}{1-\frac{z-3}{-2}} + \frac{3}{-2} \frac{1}{1-\frac{z-3}{-2}} \\
 &= \frac{z-3}{-2} \sum_{n=0}^{\infty} \left(\frac{z-3}{-2}\right)^n + \frac{3}{-2} \sum_{n=0}^{\infty} \left(\frac{z-3}{-2}\right)^n \\
 &= \sum_{n=0}^{\infty} \left(\frac{z-3}{-2}\right)^{n+1} + \sum_{n=0}^{\infty} \frac{3(z-3)^n}{(-2)^{n+1}} \\
 &= \sum_{n=1}^{\infty} \left(\frac{z-3}{-2}\right)^n + \frac{3}{-2} + \sum_{n=1}^{\infty} \frac{3}{-2} \frac{(z-3)^n}{(-2)^n} \\
 &= \frac{-3}{2} + \sum_{n=1}^{\infty} \left(1 + \frac{-3}{2}\right) \frac{(z-3)^n}{(-2)^n} \\
 &= \frac{-3}{2} + \sum_{n=1}^{\infty} \frac{-1}{2} \frac{(z-3)^n}{(-2)^n} \\
 &= \frac{-3}{2} + \sum_{n=1}^{\infty} \frac{(z-3)^n}{(-2)^{n+1}}
 \end{aligned}$$

This converges for $|z-3| < 2$

65.10 $\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \dots$ so $\sin z^2 = z^2 - \frac{z^6}{3!} + \frac{z^{10}}{5!} - \frac{z^{14}}{7!} + \dots$ and $\frac{\sin z^2}{z^4} = z^{-1} - \frac{z^2}{3!} + \frac{z^6}{5!} - \frac{z^{10}}{7!} + \dots$

$$\begin{aligned}
\frac{1}{4z - z^2} &= \frac{1}{4z} \frac{1}{1 - \frac{z}{4}} \\
&= \frac{1}{4z} \sum_{n=0}^{\infty} \left(\frac{z}{4}\right)^n \\
&= \sum_{n=0}^{\infty} \frac{z^{n-1}}{4^{n+1}} \\
&= \frac{1}{4z} + \sum_{n=0}^{\infty} \frac{z^n}{4^{n+2}}
\end{aligned}$$

Find Laurent expansions about $z_0 = 0$ for $\frac{1}{z^3-4z}$ on the regions $0 < |z| < 2$ and $|z| > 2$. When $0 < |z| < 2$ we have $\frac{1}{z^3-4z} = \frac{-1}{4z} \frac{1}{1-(\frac{z}{2})^2} = \frac{-1}{4z} \sum_{n=0}^{\infty} \left(\frac{z^2}{4}\right)^n = \sum_{n=0}^{\infty} \frac{-z^{2n-1}}{4^{n+1}}$. When $|z| > 2$, $\frac{1}{z^3-4z} = \frac{1}{z^3} \frac{1}{1-\frac{4}{z^2}} = \frac{1}{z^3} \sum_{n=0}^{\infty} \left(\frac{4}{z^2}\right)^n = \sum_{n=0}^{\infty} \frac{4^n}{z^{2n+3}}$

68.5 (just for D_2) $f(z) = \frac{-1}{(z-1)(z-2)} = \frac{1}{z-1} - \frac{1}{z-2} = \frac{1}{z} \frac{1}{1-\frac{1}{z}} - \frac{-1}{2} \frac{1}{1-\frac{z}{2}} = \frac{1}{z} \sum_{n=0}^{\infty} \frac{1}{z^n} + \frac{1}{2} \sum_{n=0}^{\infty} \frac{z^n}{2^n} = \sum_{n=0}^{\infty} \frac{1}{z^{n+1}} + \sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}} = \sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}} + \sum_{n=1}^{\infty} \frac{1}{z^n}$

Let C be the contour $|z| = 2$ oriented positively. Compute $\int_C z \cos(1/z) dz$. We have the Laurent series $z \cos(1/z) = z \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \left(\frac{1}{z}\right)^{2n} = z \cos(1/z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \frac{1}{z^{2n-1}}$. So $b_1 = -1/2$. Thus the integral is $2\pi i(-1/2) = -\pi i$.

Homework 12

72.1 Differentiating both sides of the first equation, we get $\frac{1}{(1-z)^2} = \sum_{n=1}^{\infty} n z^{n-1} = \sum_{n=0}^{\infty} (n+1) z^n$. Differentiating again, $\frac{2}{(1-z)^3} = \sum_{n=1}^{\infty} n(n+1) z^{n-1} = \sum_{n=0}^{\infty} (n+1)(n+2) z^n$. This is all valid only within the circle of convergence $|z| < 1$

72.4 Using the Taylor series for $\cos z$, which is entire, $1 - \cos z = 1 - \left(\sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!}\right) = \sum_{n=1}^{\infty} \frac{z^{2n}}{(2n)!}$. So $(1 - \cos z)/z^2 = \sum_{n=1}^{\infty} \frac{z^{2n-2}}{(2n)!}$. At $z = 0$, this is $1/2$, so this series represents $f(z)$ on all of $\mathbb{C}$. Since the series for $\cos z$ converges for all z , our new series also converges for all z (for each fixed z , multiplying by $\frac{1}{z^2}$ is multiplication by a constant and so doesn't affect the convergence). Since the series converges everywhere, it represents an entire function by the corollary in section 71.

72.6 Let C be a contour from 1 to z where z satisfies $|z-1| < 1$. Then we can integrate both sides along this contour. On the left we get $\int_C \frac{1}{w} dw = \text{Log } w|_1^z = \text{Log } z - \text{Log } 1 = \text{Log } z$. on the right we have $\int_C \sum_{n=0}^{\infty} (-1)^n (w-1)^n dw = \sum_{n=0}^{\infty} \int_C (-1)^n (w-1)^n dw = \sum_{n=0}^{\infty} (-1)^n \frac{(w-1)^{n+1}}{n+1} \Big|_1^z = \sum_{n=0}^{\infty} (-1)^n \frac{(z-1)^{n+1}}{n+1} = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(z-1)^n}{n}$.

72.7 Away from $z = 1$ on the described domain, $f(z)$ is analytic by the analyticity of Log and the quotient rule. Inside the circle $|z-1| < 1$ (and only here), $f(z) =$

$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{(z-1)^{n-1}}{n} = \sum_{n=0}^{\infty} (-1)^n \frac{(z-1)^n}{n+1}$. Note that this agrees with $f(1) = 1$, and the series absolutely converges inside the circle by comparison to the geometric series $\sum |z-1|^n$. So $f(z)$ is analytic at 1 also.

Show that if f is analytic in a domain except for finitely many singular points then they are all isolated singularities. Let $z_1, \dots, z_n$ be the singular points. Without loss of generality, consider z_1 . For each $i > 1$, let $R_i = |z_1 - z_i|$. Since there are finitely many singular points, there is a minimum $R = \min\{R_2, \dots, R_n\}$. Now choose ϵ so that $0 < \epsilon < R$ and so that z_1 has an ϵ neighborhood in the domain (this is possible because domains are open). Then f is analytic on the deleted ϵ neighborhood around z_1 , so z_1 is an isolated singular point. The argument is the same for the other singular points.

77.1.a $\frac{1}{z+z^2} = \frac{1}{z} \frac{1}{1+z} = \frac{1}{z}(1 - z + \dots) = \frac{1}{z} - 1 + \dots$ for $|z| < 1$. So the residue at 0 is 1.

77.1.b $z \cos(1/z) = z(1 - \frac{1}{2z^2} + \dots) = z - \frac{1}{2z} + \dots$ on the whole plane. So the residue is $-1/2$.

77.1.c $\frac{z - \sin z}{z} = \frac{1}{z}(z - (z - z^3/6 + \dots)) = \frac{1}{z}(z^3/6 + \dots) = z^2/6 + \dots$. So the residue is 0.

77.2.a $\frac{e^{-z}}{z^2} = \frac{1}{z^2}(1 - z + z^2/2 + \dots) = \frac{1/z^2}{-} \frac{1}{z} + \dots$. So the integral is $2\pi i(-1) = -2\pi i$.

77.2.d There are singularities at 0 and 2. By partial fractions, $\frac{z+1}{z^2-2z} = \frac{-1}{2z} + \frac{3}{2(z-2)}$. At $z = 0$, the second function is analytic, so the residue is $-1/2$. At $z = 2$, the first function is analytic so the residue is $3/2$. So the integral is $2\pi i(-1/2 + 3/2) = 2\pi i$.

Let C be the positively oriented circle $|z| = 5$. Compute $\int_C \frac{\sin z}{(z-\pi)^2}$. To find the residue at π , we need to expand $\sin z$ in powers of $z - \pi$. We use $\sin(z) = \sin(z - \pi + \pi) = \sin(z - \pi) \cos \pi + \cos(z - \pi) \sin \pi = -\sin(z - \pi) = -(z - \pi) + (z - \pi)^3/6 - \dots$. So $\frac{\sin z}{(z-\pi)^2} = -1/(z - \pi) + \dots$. So the residue is -1 and the integral is $-2\pi i$.

77.3 Looking at $\frac{f(1/z)}{z^2}$ we have $\frac{1}{z^2} \frac{4/z-5}{(1/z)(1/z-1)} = \frac{1}{z^2} \frac{z(4-5z)}{1-z} = \frac{4-5z}{z(1-z)}$. The residue at 0 is $\frac{4-5(0)}{(1-0)} = 4$, so the integral is $8\pi i$.

77.4.a $\frac{1}{z^2} f(1/z) = \frac{1}{z^2} \frac{1/z^5}{1-1/z^3} = \frac{1}{z^7} \frac{1}{1-1/z^3} = \frac{1}{z^4} \frac{-1}{1-z^3} = \frac{-1}{z^4} \sum_{n=0}^{\infty} z^{3n}$. So the residue is -1 and the integral is $-2\pi i$.

77.4.b $\frac{1}{z^2} f(1/z) = \frac{1}{z^2} \frac{1}{1+1/z^2} = \frac{1}{z^2+1}$. This is analytic at 0 so the residue at 0 is 0. So the integral is 0.

77.7 Consider $P(1/z)/Q(1/z)$. The biggest power of $1/z$ in the expression is $1/z^m$. Multiplying top and bottom by z^m , we get a polynomial with non-zero constant term on the bottom and a polynomial with 0 constant term and linear term on the top. Dividing by z^2 , the numerator is still a polynomial and the denominator is a polynomial with non-zero constant term. So $\frac{1}{z^2} \frac{P(1/z)}{Q(1/z)}$ is analytic at 0, so its residue at 0 is 0 and the integral is 0.